

Deep Hedging

MA529/DS595: Stochastic Processes for Generative Deep Learning

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1. Introduction and Problem Statement

Derivative hedging is a cornerstone of modern financial risk management. In a textbook frictionless market — continuous trading, no bid-ask spreads, unlimited liquidity — the Black-Scholes framework provides an exact replication strategy: the famous delta hedge. Yet real trading desks operate in a world of market frictions that systematically violate these idealized assumptions.

The paper by Buehler, Gonon, Teichmann and Wood (2019) confronts this gap directly. Their central observation is simple but powerful: when transaction costs, market impact, liquidity constraints, and risk limits are present, classical pricing and hedging theory breaks down entirely. Pricing and risk are no longer linear quantities. The optimal strategy depends not only on the instrument being hedged but on the trader's risk preferences and the composition of the entire portfolio — a feature practitioners call having an "axe."

The authors propose Deep Hedging: a model-free reinforcement learning framework that learns optimal hedging strategies by training neural networks to minimize a convex risk measure over simulated market scenarios. The framework requires no Greeks, no equivalent martingale measure, and no closed-form derivative pricing formula. It only requires sample paths of the hedging instruments and the payoff of the liability. This makes it directly applicable to over-the-counter derivatives for which no observable market price exists.

The practical importance is significant. Computing the exponential utility indifference price for even a simple European call under proportional transaction costs in Black-Scholes requires solving a multidimensional nonlinear free boundary problem. The deep hedging algorithm sidesteps this entirely, producing close-to-optimal hedging strategies efficiently in both low- and high-dimensional settings.

2. Mathematical Framework

2.1 Discrete-Time Market with Frictions

The market is modeled on a finite probability space $\Omega = \{\omega_1, \dots, \omega_N\}$ with a probability measure P , over a discrete time horizon

with trading dates $0 = t_0 < t_1 < \dots < t_n = T$. At each time step t_k , the agent observes market information I_k (prices, volatilities, news, risk limits, etc.), generating a filtration $F = (F_k)$. The d hedging instruments have mid-prices $S = (S_k)$, which need not form a martingale under P .

A trading strategy $\delta = (\delta_k)$ is an F -adapted process where $\delta_k \in \mathbb{R}^d$ denotes holdings in each instrument at time t_k . Strategies are subject to constraints H_k capturing liquidity restrictions (e.g., an option that cannot be traded before it is listed). Trading incurs costs: $CT(\delta) = \sum_k c_k(\delta_k - \delta_{k-1})$ where $c_k(n)$ is an adapted cost function. For proportional transaction costs, $c_k(n) = \varepsilon \cdot \sum_i |n^i| \cdot S_{ik}$. The agent's terminal P&L given initial cash p_0 and liability Z is: $PLT(Z, p_0, \delta) = -Z + p_0 + (\delta \cdot S)_T - CT(\delta)$.

2.2 Convex Risk Measures

The optimality criterion is a convex risk measure $\rho: X \rightarrow \mathbb{R}$ satisfying three axioms:

- Monotonicity: $X_1 \geq X_2 \implies \rho(X_1) \leq \rho(X_2)$. A better position is less risky.
- Convexity: $\rho(\alpha X_1 + (1-\alpha)X_2) \leq \alpha \rho(X_1) + (1-\alpha)\rho(X_2)$. Diversification is rewarded.
- Cash-invariance: $\rho(X + c) = \rho(X) - c$. Adding cash reduces risk by exactly that amount.

Cash-invariance has a crucial interpretation: $\rho(X)$ is the minimal cash injection needed to make position X acceptable (i.e., $\rho(X + \rho(X)) = 0$). Two primary examples appear in the paper:

OCE risk measures (Optimized Certainty Equivalents): $\rho(X) = \inf_{w \in \mathbb{R}} \{w + E[\ell(-X-w)]\}$, where ℓ is a convex loss function. The Average Value-at-Risk (CVaR/Expected Shortfall) at level α is obtained by setting $\ell(x) = (1-\alpha)^{-1} \max(x, 0)$.

Entropic risk measure: $\rho(X) = \lambda^{-1} \log E[\exp(-\lambda X)]$, which corresponds to exponential utility indifference pricing with risk aversion parameter $\lambda > 0$.

2.3 Optimal Hedging Problem and Indifference Price

Given the risk measure ρ , the optimal hedging value is: $\pi(X) = \inf_{\delta \in H} \rho(X + (\delta \cdot S)_T - CT(\delta))$

Proposition 3.2 establishes that π is itself a convex risk measure when $CT(\cdot)$ and H are convex. The indifference price of liability Z — the cash amount making the agent indifferent between hedging and not hedging — is defined as: $p(Z) = \pi(-Z) - \pi(0)$

This is the minimal premium an agent must charge for liability Z , beyond what they would charge on an empty book, to remain equally well off. In a frictionless complete market this reduces to the classical replication price (Lemma 3.3), and $p(Z) \geq E[Z]$ for OCE measures (Proposition 3.10).

3. The Deep Hedging Algorithm

3.1 Neural Network Parameterization of Strategies

The key idea is to replace the infinite-dimensional optimization over all adapted strategies $\delta \in H$ with a finite-dimensional optimization over neural network parameters θ . At each time step t_k , the holdings are given by a feedforward neural network: $\delta_k = F_k^{\theta}(I_0, \dots, I_k, \delta_{k-1})$ where F_k^{θ} is a neural network with ReLU activations ($\sigma(x) = \max(x, 0)$) and two hidden layers. The recurrent dependence on δ_{k-1} is crucial: it allows the network to incorporate the current portfolio position into its decision, which matters especially when transaction costs penalize large changes in holdings.

Each network F_k^{θ} maps from $\mathbb{R}^{r(k+1)+d} \rightarrow \mathbb{R}^d$, where r is the dimension of the market information vector at each step. The parameters $\theta = (\theta_0, \dots, \theta_{n-1})$ are different for each time step. The set of neural network strategies HM (networks with at most M non-zero weights) approximates the full strategy space H as $M \rightarrow \infty$.

3.2 Training: Stochastic Gradient Descent

For OCE risk measures, substituting the neural network strategies into the optimization problem yields: $\pi_M(-Z) = \inf_{\{\theta, w\}} \{ w + E[\ell(Z - (\delta^{\theta} \cdot S)T + CT(\delta^{\theta}) - w)] \}$. This is a finite-dimensional optimization problem in parameters $(w, \theta) \in \mathbb{R} \times \Theta_M$. Three ingredients make training practical:

- Mini-batch SGD: At each gradient step, the expectation over the (potentially huge) sample space Ω is replaced by an average over a random mini-batch of scenarios. This reduces computational cost from $O(N)$ to $O(N_{\text{batch}})$ per step, and the randomness helps escape local minima.
- Training uses Adam (Kingma & Ba, 2015) with backpropagation for efficient gradient computation.

The training objective $J(\theta) = w + E[\ell(Z - (\delta^{\theta} \cdot S)T + CT(\delta^{\theta}) - w)]$ decomposes as a sum over sample paths, enabling parallel computation. The Adam algorithm with learning rate 0.005 and batch size 256 was used in the paper's experiments. Batch normalization in each hidden layer allows larger learning rates.

3.3 Extension to General Convex Risk Measures

For risk measures beyond the OCE class, the paper appeals to the robust representation theorem (Theorem 4.12): any convex risk measure on a finite probability space can be written as: $\rho(X) = \max_{Q \in P} \{ E_Q[-X] - \alpha(Q) \}$ where P is the set of probability measures on Ω and $\alpha(Q)$ is the penalty function. Theorem 4.13 shows that the supremum over probability measures can itself be approximated by neural networks parameterized by log-likelihood functions. This converts the hedging problem into a minimax problem: $\inf_{\theta_0} \sup_{\theta_1} J(\theta_0, \theta_1)$ where θ_0 parameterizes the hedging strategy and θ_1 parameterizes the adversarial

probability measure. Both inner optimizations satisfy the properties needed for stochastic gradient descent (gradient decomposes over samples; gradients are computable via backpropagation). This extends deep hedging to coherent risk measures such as CVaR.

4. Theoretical Justification: Why It Works

4.1 Universal Approximation

The theoretical foundation rests on the Universal Approximation Theorem (Hornik, 1991): for any bounded, non-constant activation function σ , the set of neural networks $\text{NN}\sigma_{\{\infty, d_0, 1\}}$ is dense in $L^p(\mathbb{R}^{d_0}, \mu)$ for any finite measure μ and $p \in [1, \infty)$.

The intuition is geometric: a neural network with one hidden layer creates a linear combination of shifted and scaled copies of σ . As more neurons are added, these basis functions can approximate any sufficiently regular function arbitrarily well in L^p . This is the sense in which neural networks are "universal" function approximators — not that any fixed architecture suffices, but that by growing the network we can get arbitrarily close to any target function.

In the deep hedging context, each component of the optimal hedging strategy $\delta_k(\omega)$ is a measurable function of the market information $(I_0, \dots, I_k)(\omega)$. Since Ω is finite, these functions are bounded and lie in $L^1(P)$. The Universal Approximation Theorem therefore guarantees they can be approximated by neural networks to arbitrary precision.

4.2 Convergence of Neural Network Prices (Proposition 4.9)

The central theoretical result is Proposition 4.9: as the network size $M \rightarrow \infty$, $\pi_M(X) \rightarrow \pi(X)$ for all $X \in X$.

The proof has a clean structure that illuminates why the algorithm works:

Step 1 (Monotone approximation from above): Since $\text{HM} \subseteq \text{HM}+1 \subseteq H^u$, restricting the optimization to larger network classes yields smaller values: $\pi_M(X) \geq \pi_{M+1}(X) \geq \pi(X)$. The neural network price is always an upper bound on the true optimal price, and it decreases monotonically as networks grow.

Step 2 (Approximation from below): Fix $\varepsilon > 0$. The infimum in $\pi(X)$ is (approximately) achieved by some strategy δ . By the Universal Approximation Theorem, neural networks $F_{k,n}$ can be found such that $F_{k,n}(I_0, \dots, I_k) \rightarrow \delta_k$ pointwise on the finite Ω . By continuity of ρ (which holds since Ω is finite, making ρ a finite-dimensional convex function) and upper semi-continuity of the cost functions c_k , the risk measure evaluated on the neural network strategies converges to the risk measure evaluated on δ . So eventually $\pi_M(X) \leq \pi(X) + \varepsilon$.

The finiteness of Ω plays a dual role: it ensures (a) every F_k -measurable random variable has a measurable representation as a function of $(I_0, \dots, I_k)$, enabling the UAT to apply; and (b) ρ is a continuous finite-dimensional convex function, enabling continuity arguments in the convergence proof.

4.3 Geometric Intuition

The convergence can be visualized as follows. The true optimal hedging strategy lives in an infinite-dimensional function space. The neural network strategies HM form a growing sequence of finite-dimensional subsets, like successively finer grids covering the space. Proposition 4.9 says that the optimal value of the risk measure objective on these finite-dimensional subsets converges to the global optimum — the mesh of the approximating grid becomes fine enough that minimizing over it is nearly the same as minimizing over the full space.

The convexity of ρ is what makes this work smoothly: small perturbations in the strategy δ (from the true optimum to its neural network approximation) lead to small changes in the objective $\rho(X + (\delta \cdot S)T - CT(\delta))$, because ρ is continuous on the finite sample space.

5. Numerical Experiments

The paper validates the algorithm under the Heston stochastic volatility model, with stock price S^1 and variance swap S^2 as hedging instruments ($d = 2$). Time horizon: 30 trading days with daily rebalancing.

5.1 Benchmark: No Transaction Costs

Under the Heston model with CVaR ($\alpha = 0.5$), the deep hedging strategy closely matches the classical model delta hedge. More risk-averse settings ($\alpha = 0.99$) produce higher prices ($p_0 = 3.49$ vs. risk-neutral 1.69) and strategies that protect more aggressively against tail losses, consistent with theory.

5.2 Price Asymptotics Under Transaction Costs

For small proportional transaction costs ε , the known theoretical result for Black-Scholes is $p_\varepsilon - p_0 = O(\varepsilon^{2/3})$. The deep hedging algorithm numerically reproduces this scaling (observed rate ≈ 0.67), validating the method even where no closed-form formula is available for comparison. Applied to the Heston model with $d = 2$ instruments — a case with no prior theoretical results — the algorithm produces a rate of ≈ 0.71 , suggesting the $2/3$ scaling is universal across a broader class of models.

5.3 High-Dimensional Scalability

A high-dimensional experiment with $d = 10$ hedging instruments (5 independent Heston models) demonstrates scalability. For variance-optimal hedging, the sum-of-calls payoff $Z = \sum Z_h$ decomposes by independence, so the optimal strategy is decoupled across sub-models. The network recovers this decoupled structure, achieving $\sim 5\times$ the single-model loss as predicted. Training time scales modestly (5.75h for $d=10$ vs. 2.1h for $d=2$ on a standard laptop), showcasing the computational efficiency of gradient-based deep learning.

6. Conclusions

Deep hedging represents a genuine paradigm shift in derivative risk management. Its key contributions are:

- Model-freedom: No equivalent martingale measure, no Greeks, no closed-form pricing formula required — only sample paths.
- Theoretical rigor: Proposition 4.9 guarantees convergence to the true optimal hedging strategy as network size grows, grounded in the Universal Approximation Theorem.
- Generality: Handles arbitrary convex risk measures, market frictions, liquidity constraints, and high-dimensional portfolios within a single unified framework.
- Empirical validation: Recovers the classical delta hedge in frictionless markets, the known $\varepsilon^{2/3}$ price scaling under transaction costs, and demonstrates tractability at $d = 10$ instruments.

The deepest mathematical insight is that convexity is the load-bearing structure throughout: convexity of the risk measure ρ enables Proposition 3.2 (π is a risk measure), Proposition 4.9 (convergence via continuity of ρ), and the gradient-based training (the objective is convex-ish in a favorable way). The neural network is not just a black box — it is the right function class because the Universal Approximation Theorem guarantees it can represent any measurable strategy, and because its compositional structure makes gradient computation via backpropagation efficient.

Future directions noted by the authors include systematic treatment of trading constraints via Lagrange multipliers, extension to dynamic risk measures in a truly multi-period sense, and more realistic market impact models. The framework has since become highly influential in quantitative finance, with extensions to path-dependent payoffs, rough volatility models, and online learning settings.

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